



Barker
College

2021

YEAR 11
PRELIMINARY EXAMINATION

Mathematics Advanced

Staff Involved:

- HYB: H Bawa • PCB: P Boyages • AXD: A Davis • LMD: L de Gorter
- GPF: G Fitzmaurice • GDH: G Hanlon • KJL: K Lee • ESP: E Pratt
- RAS: R Smith* • JZT: J Thomas • ALY: A Young

TOPICS COVERED:

Algebra, Number and Surds	Exponential and Logarithmic Functions
Functions and Graphs	Differentiation and Curve Sketching
Transformations and Symmetry	Extending Calculus
The Coordinate Plane	Probability
Trigonometry	

205 copies

Friday 3 September, 8:30am

General

Instructions:

- Reading Time – 5 minutes
- Working Time - 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- Reference sheet is provided separately and at the end of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale unless specifically stated

Total marks:

80

Section I – 10 marks (pages 2-5)

- Use Canvas to select your preferred response for Questions 1-10
- Allow about 15 minutes for this section

Section II – 70 marks (pages 6-14)

- Attempt Questions 11-14
- Allow about 1 hour and 45 minutes for this section
- Answer the questions on your own lined paper, clearly indicating which question you are answering
- Write your student number on all answer pages
- Section II will be uploaded in 4 sections: Q11, Q12, Q13, Q14

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use Canvas to select your preferred response for Questions 1 – 10.

1. Which of the following is the value of $|5 - 7| + |-6| - |-5| + 2|-3|$?

- A. -9
- B. -7
- C. 9
- D. 19

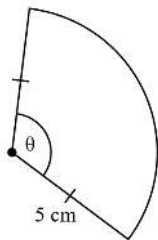
2. Which of the following is the expansion of $(5 - 2\sqrt{3})^2$?

- A. -11
- B. $25 - 4\sqrt{3}$
- C. 37
- D. $37 - 20\sqrt{3}$

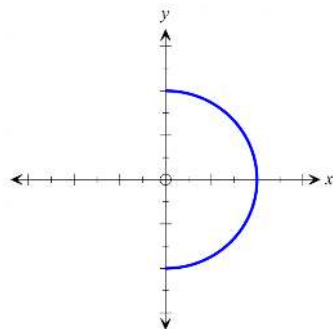
3. What is the *domain* of the function $y = \sqrt{5x - 2}$?

- A. $[0, \infty)$
- B. $[\frac{2}{5}, \infty)$
- C. $(\frac{2}{5}, \infty)$
- D. $(\infty, \frac{2}{5})$

4. The following sector has an area of $10\pi \text{ cm}^2$.
What is the size of the angle θ , subtended at its centre?



- A. $\frac{\pi}{5}$
B. $\frac{2\pi}{5}$
C. $\frac{4\pi}{5}$
D. 2π
5. What type of relation is the following graph?



- A. One-to-one
B. One-to-many
C. Many-to-one
D. Many-to-many

6. What is the angle of inclination, created with the x -axis, of the line $4x - 5y + 30 = 0$?

- A. $38^\circ 40'$
B. $51^\circ 20'$
C. $75^\circ 58'$
D. $141^\circ 20'$

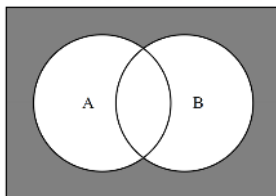
7. The function $y = x^2$ is translated 3 units to the right and 5 units up.
What is its new equation?

- A. $y = x^2 + 3x + 5$
B. $y = x^2 - 6x + 14$
C. $y = x^2 - 9x + 5$
D. $y = x^2 + 9x + 14$

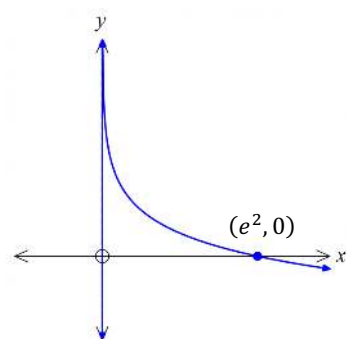
8. If $\log_x y = 2\log_x a + \log_x \left(\frac{b}{a}\right) - \log_x b$, then the value of y is:

- A. a
B. ab
C. a^2b
D. 1

9. A Venn diagram is shaded as per below.
Which of the following represents the shaded area in set notation?



- A. $A \cup B$
B. $A \cap B$
C. $\bar{A} \cup \bar{B}$
D. $\bar{A} \cap \bar{B}$
10. The graph of the function $y = a \ln x + b$ is shown below.
What are the values of a and b ?



- A. $a = 1, b = e^2$
B. $a = 1, b = 0$
C. $a = -1, b = 2$
D. $a = -1, b = e$

End of Section I

Section II

70 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

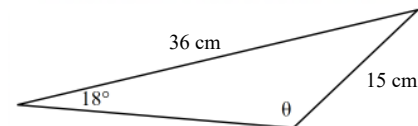
Write your student number on all answer pages.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

START A NEW PAGE SINCE QUESTION 11 WILL BE UPLOADED SEPARATELY

Question 11 (17 marks)

- a. Convert 5 radians into degrees and minutes, rounded to the nearest minute. 2
- b. Solve $|4x - 9| = 15$. 2
- c. Find the value of the obtuse angle θ in the triangle below. Round your answer to the nearest minute. 2



d. If $f(x) = \frac{3}{x^2}$, find an expression for $f(f(x))$. 2

e. Given $\log_x 4 \approx 1.26$ and $\log_x 6 \approx 1.63$, evaluate:

i. $\log_x 24$ 1

ii. $\log_x 9$ 2

f. The population of a small town is given by $P = 20\,000e^{kt}$, where t is the time in years after 1st January, 1900. After 5 years, the town's population had halved.

i. Find the initial population. 1

ii. Calculate the value of the constant k to 4 decimal places. 2

iii. What was the population in 1920? 1

iv. In which year did the population fall below 100 residents? 2

START A NEW PAGE SINCE QUESTION 12 WILL BE UPLOADED SEPARATELY

Question 12 (17 marks)

a. During a particular month, the probability of it raining on a given day is $\frac{1}{3}$.
Over a three-day long weekend, what is the probability that it will rain on exactly two days? 2

b. Differentiate the following. Leave your answers with a positive, non-fractional index, where necessary.

a. $y = 4e^{3x} + 6$ 1

β. $y = (3x^2 + 5x)^4$ 1

γ. $f(x) = 5x^{-2} - 6x^{\frac{1}{2}}$ 2

δ. $f(x) = 3xe^{2x}$ 2

c. Find the value of $\sin \frac{\pi}{3} + \sec \frac{\pi}{4}$.
Leave your answer as a single fraction in exact form. 2

End of Question 11

- d. Find the value(s) of k for which $y = 3x^2 - 12x + k$ would have two different roots. 2

- e. Determine the transformation that has been applied to $x^2 + y^2 = 49$ to result in $x^2 + y^2 - 6y - 40 = 0$. 2

- f. Sketch the function $f(x) = 1 - \frac{1}{x-2}$, showing all intercepts and other important features. 3

START A NEW PAGE SINCE QUESTION 13 WILL BE UPLOADED SEPARATELY

Question 13 (18 marks)

- a. Differentiate $f(x) = \frac{5x^3}{2x-1}$ 2

- b. Consider the curve $y = x^3 - px^2 + 2x + 5$. At the point where $x = 2$, the tangent to this curve is parallel to the line $y = -6x + 3$. Find the value of p . 3

- c. In a class of 25 students, 18 play netball, 13 play soccer and 2 play neither sport.

- i. Draw a Venn diagram for this scenario, showing all information. 2

- ii. A student is chosen at random. Find the probability that:

- α. They play both soccer and netball. 1

- β. They do not play soccer. 1

- d. Find the exact equation of the straight line which passes through the midpoint of (0, -5) and (4, -1) and is perpendicular to the line that makes an angle of 30° with the positive x -axis. 3

End of Question 12

- e. For the function $y = -(x + 2)(x - 1)^2$:
- i. Graph the function, showing x and y -intercepts. 2
- ii. Show very clearly, without skipping steps, why $\frac{dy}{dx} = 3(1 - x)(x + 1)$. 2
- iii. Find the coordinates of any stationary points and add them to your graph. 2

Note that if these stationary points do not seem to fit your graph from i., this indicates a mistake somewhere. As you try to resolve the mistake, be careful not to compound the error by altering anything that is correct.

End of Question 13

START A NEW PAGE SINCE QUESTION 14 WILL BE UPLOADED SEPARATELY

Question 14 (18 marks)

- a. Three sets are defined as below.

Universal Set $V = \{\text{Positive integers} \leq 20\}$

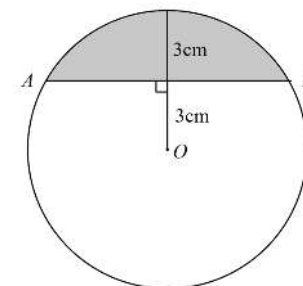
Subset $A = \{\text{Even numbers}\}$

Subset $B = \{\text{Multiples of 3}\}$ Note that 3 is a multiple of 3.

A number is randomly selected from Set V . Calculate:

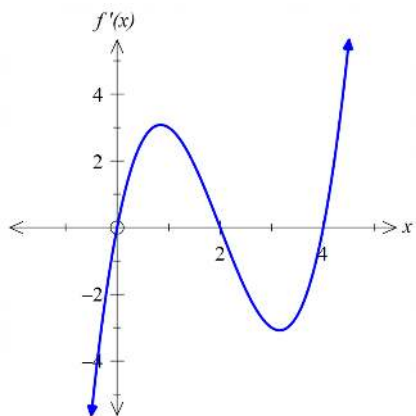
- i. $P(A)$ 1
- ii. $P(B)$ 1
- iii. $P(A \cap B)$ 1
- iv. $P(A \cup B)$ 1

- b. A circle has a radius of 6cm and centre O . A chord AB has been drawn, such that it bisects the radius, as shown below.



- i. Show the size of $\angle AOB$ is $\frac{2\pi}{3}$ radians. 2
- ii. Calculate the area of the shaded minor segment to 2 decimal places. 3

- c. A function $y = f(x)$ has its **gradient function** $y = f'(x)$ graphed as below.



- i. For which x value(s) does the original function $f(x)$ have a horizontal tangent? 1
- ii. For what values of x is the function $f(x)$ decreasing? 1

- d. Show that $\left(\operatorname{cosec} x + \frac{\cos x}{\sin x}\right) \left(\operatorname{cosec} x - \frac{\cos x}{\sin x}\right) = 1$. 2

- e. Let $f(x) = \ln x$, and $g(x) = x^4 + 1$.
- i. Find $h(x)$ if $h(x) = f(g(x))$. 1
- ii. State the domain and range of $h(x)$. 1
- iii. State whether the functions $f(x)$, $g(x)$ and $h(x)$ are even, odd or neither. 1
- iv. Show that $h(x) + h(-x) = f\left((g(x))^2\right)$. 1
- v. Let $i(x) = g(f(x))$.
By first determining whether or not $i(x) = 4\ln x + 1$, state the range of $i(x)$. 1

End of Paper

Year 11 Mathematics Advanced - Student Solutions

Section I: Multiple Choice

$$1. |5-7| + |-6| - |-5| + 2|-3|$$

$$= |-2| + 6 - 5 + 2 \times 3$$

$$= 2 + 6 - 5 + 6$$

$$= 9 \quad \boxed{C}$$

$$2. (5-2\sqrt{5})^2$$

$$= (5-2\sqrt{5})(5-2\sqrt{5})$$

$$= 25 - 20\sqrt{5} + 12$$

$$= 37 - 20\sqrt{5} \quad \boxed{D}$$

$$3. \text{Domain of } y = \sqrt{5x-2}$$

$$5x-2 \geq 0$$

$$x \geq \frac{2}{5}$$

$$\left[\frac{2}{5}, \infty\right) \quad \boxed{B}$$

$$4. A = \frac{1}{2}r^2\theta$$

$$10\pi = \frac{1}{2} \times 5^2 \times \theta$$

$$\theta = \frac{4\pi}{5} \quad \boxed{C}$$

$$5. \text{One-to-many} \quad \boxed{B}$$

$$6. 4x - 5y + 30 = 0$$

$$5y = 4x + 30$$

$$y = \frac{4}{5}x + 6$$

$$m = \frac{4}{5}$$

$$\frac{4}{5} = \tan \theta$$

$$\theta = \tan^{-1}\left(\frac{4}{5}\right)$$

$$\theta = 38^\circ 40' \quad \boxed{A}$$

$$7. y = x^2 \Rightarrow y = (x-3)^2 + 5$$

$$y = (x-3)(x-3) + 5$$

$$y = x^2 - 6x + 9 + 5$$

$$y = x^2 - 6x + 14 \quad \boxed{B}$$

$$8. \log_x y = 2 \log_x a + \log_x \left(\frac{b}{a}\right) - \log_x b$$

$$\log_x y = \log_x a^2 + \log_x \left(\frac{b}{a}\right) - \log_x b$$

$$\log_x y = \log_x \left(\frac{a^2 b}{a}\right) - \log_x b$$

$$\log_x y = \log_x \left(\frac{a^2 b}{ab}\right)$$

$$\log_x y = \log_x a$$

$$\therefore y = a \quad \boxed{A}$$

$$9. \bar{A} \cap \bar{B} \quad \boxed{D}$$

$$10. a = -1$$

$$\text{so } y = -\ln x + b$$

$$\text{The curve passes through } (e^2, 0)$$

$$0 = -\ln e^2 + b$$

$$0 = -2 \ln e + b$$

$$0 = -2 + b$$

$$b = 2 \quad \boxed{C}$$

Section II: Question 11

$$a) 1 \text{ radian} = \frac{180^\circ}{\pi}$$

$$5 \text{ radians} = 5 \times \frac{180^\circ}{\pi}$$

$$= 286^\circ 29'$$

$$b) |4x-9| = 15$$

$$4x-9 = 15 \quad \text{or} \quad 4x-9 = -15$$

$$4x = 24 \quad \quad \quad 4x = -6$$

$$x = 6 \quad \quad \quad x = -\frac{3}{2}$$

$$c) \frac{\sin \theta}{36} = \frac{\sin 18}{15}$$

$$\sin \theta = \frac{36 \sin 18}{15}$$

$$\sin \theta = 0.7416 \dots$$

$$\theta = 47.87^\circ, 180 - 47.87^\circ$$

$$\theta = 47.87^\circ, 132.1286^\circ$$

$$\text{Since } \theta \text{ is obtuse}$$

$$\theta = 132^\circ 8'$$

$$d) f(f(x)) = \frac{3}{\left(\frac{3}{x}\right)^2}$$

$$= \frac{3}{\frac{9}{x^2}}$$

$$= \frac{3x^2}{9}$$

$$= \frac{x^2}{3}$$

$$e) i) \log_x 24 = \log_x 4 + \log_x 6$$

$$\approx 1.26 + 1.63$$

$$\approx 2.89$$

$$ii) \log_x 9 = \log_x \left(\frac{36}{4}\right)$$

$$= \log_x 36 - \log_x 4$$

$$= \log_x 6^2 - \log_x 4$$

$$= 2 \log_x 6 - \log_x 4$$

$$\approx 2 \times 1.63 - 1.26$$

$$\approx 2$$

$$f) i) P = ?, \text{ when } t = 0$$

$$P = 20000e^{k \cdot 0}$$

$$P = 20000e^0$$

$$P = 20000$$

$$ii) P = 10000 \text{ when } t = 5$$

$$10000 = 20000e^{5k}$$

$$\frac{10000}{20000} = e^{5k}$$

$$\frac{1}{2} = e^{5k}$$

$$\ln\left(\frac{1}{2}\right) = 5k$$

$$k = \frac{1}{5} \ln\left(\frac{1}{2}\right)$$

$$k = -0.1386$$

$$iii) P = ?, t = 20$$

$$P = 20000e^{-0.1386 \times 20}$$

$$P = 1250$$

$$iv) P = 100, t = ?$$

$$100 = 20000e^{-0.1386t}$$

$$\frac{100}{20000} = e^{-0.1386t}$$

$$\ln\left(\frac{100}{20000}\right) = -0.1386t$$

$$t = \ln\left(\frac{100}{20000}\right) \div -0.1386$$

$$t = 38.2 \dots$$

$\therefore$ The population fell below 100 residents in 1938.

Question 12

a) Probability of rain on two days

$$P(RR'R') = \frac{1}{3} \times \frac{1}{3} \times \frac{2}{3}$$

$$P(RR'R') = \frac{2}{27}$$

$$P(R'R'RR) = \frac{2}{27} \times \frac{1}{3} \times \frac{1}{3}$$

$\therefore$ Probability of rain on exactly two days

$$\frac{2}{27} + \frac{2}{27} + \frac{2}{27} \text{ or } 3 \times \frac{2}{27}$$

$$b) \alpha. y = 4e^{3x} + 6$$

$$\frac{dy}{dx} = 3 \times 4e^{3x} + 0$$

$$\frac{dy}{dx} = 12e^{3x}$$

$$\beta. y = (3x^2 + 5x)^4$$

$$\frac{dy}{dx} = 4 \times (3x^2 + 5x)^{4-1} \times (6x + 5)$$

$$\frac{dy}{dx} = 4(6x + 5)(3x^2 + 5x)^3$$

$$\gamma. f(x) = 5x^{-2} - 6x^{\frac{1}{2}}$$

$$f'(x) = -2 \times 5x^{-3} - \frac{1}{2} \times 6x^{-\frac{1}{2}}$$

$$f'(x) = -10x^{-3} - 3x^{-\frac{1}{2}}$$

$$f(x) = -\frac{10}{x^3} - \frac{3}{\sqrt{x}}$$

$$\delta. f(x) = 3xe^{2x}$$

$$u = 3x \quad v = e^{2x}$$

$$u' = 3 \quad v' = 2e^{2x}$$

$$f'(x) = 3 \times e^{2x} + 2e^{2x} \times 3x$$

$$f'(x) = 3e^{2x} + 6xe^{2x} \text{ or } 3e^{2x}(1 + 2x)$$

$$c) \sin\left(\frac{\pi}{3}\right) + \sec\left(\frac{\pi}{4}\right)$$

$$= \sin\left(\frac{\pi}{3}\right) + \frac{1}{\cos\left(\frac{\pi}{4}\right)}$$

$$= \frac{\sqrt{3}}{2} + \frac{1}{\frac{1}{\sqrt{2}}}$$

$$= \frac{\sqrt{3}}{2} + \sqrt{2}$$

$$= \frac{\sqrt{3} + 2\sqrt{2}}{2}$$

d) $y = 3x^2 - 12x + k$ has two different roots

When $b^2 - 4ac > 0$

$$(-12)^2 - 4(3)(k) > 0$$

$$144 - 12k > 0$$

$$144 > 12k$$

$$k < 12$$

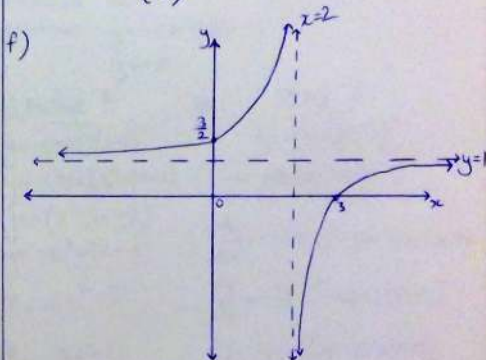
$$e) x^2 + y^2 = 49 \rightarrow x^2 + y^2 - 6y - 40 = 0$$

$$x^2 + (y-3)^2 - 3^2 - 40 = 0$$

$$x^2 + (y-3)^2 = 49$$

Vertical translation of 3 units up

Translation $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$



$$f(x) = 1 - \frac{1}{x-2}$$

Vertical asymptote: $x = 2$

Horizontal asymptote: $y = 1$

$$y\text{-intercept: } f(0) = 1 - \frac{1}{0-2}$$

$$= 1 + \frac{1}{2}$$

$$= \frac{3}{2}$$

$$x\text{-intercept: } 0 = 1 - \frac{1}{x-2}$$

$$\frac{1}{x-2} = 1$$

$$x-2 = 1$$

$$x = 3$$

Question 13

$$a) f(x) = \frac{5x^3}{2x-1}$$

$$u = 5x^3 \quad v = 2x-1$$

$$u' = 15x^2 \quad v' = 2$$

$$v' = (2x-1)^2$$

$$f'(x) = \frac{15x^2(2x-1) - 2 \times 5x^3}{(2x-1)^2}$$

$$f'(x) = \frac{30x^3 - 15x^2 - 10x^3}{(2x-1)^2}$$

$$f'(x) = \frac{20x^3 - 15x^2}{(2x-1)^2} \text{ or } \frac{5x^2(4x-3)}{(2x-1)^2}$$

$$b) \frac{dy}{dx} = -6 \text{ when } x=2$$

$$y = x^3 - px^2 + 2x + 5$$

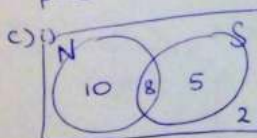
$$\frac{dy}{dx} = 3x^2 - 2px + 2$$

$$-6 = 3(2)^2 - 2p \times 2 + 2$$

$$-6 = 12 - 4p + 2$$

$$4p = 20$$

$$p = 5$$



$$ii) a. \frac{8}{25} \quad \phi. \frac{12}{25}$$

$$d) m = \tan 30$$

$$m = \frac{1}{\sqrt{3}}$$

$$\text{Perpendicular} = -\sqrt{3}$$

$$\text{Midpoint } \left(\frac{0+4}{2}, \frac{-5+1}{2}\right)$$

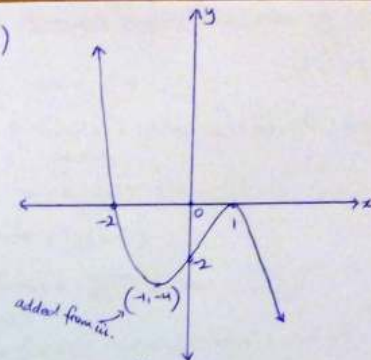
$$(2, -3)$$

$$y - 3 = -\sqrt{3}(x - 2)$$

$$y + 3 = -x\sqrt{3} + 2\sqrt{3}$$

$$y = -x\sqrt{3} + 2\sqrt{3} - 3$$

e) i)



$$y = -(x+2)(x-1)^2$$

Shape: $\curvearrowright$ bounce

$$x\text{-intercepts: } -2, 1$$

$$y\text{-intercept: } y = -(0+2)(0-1)^2$$

$$y = -2$$

ii) Method A

$$y = -(x+2)(x-1)^2$$

$$y = -(x+2)(x^2 - 2x + 1)$$

$$y = -(x^3 - 3x^2 + 2x - 2)$$

$$y = -x^3 + 3x^2 - 2x + 2$$

$$\frac{dy}{dx} = -3x^2 + 6x - 2$$

$$\frac{dy}{dx} = -3(x^2 - 2x + 1)$$

$$\frac{dy}{dx} = -3(x-1)(x-1)$$

$$\frac{dy}{dx} = 3(1-x)(x-1)$$

$$\frac{dy}{dx} = 3(1-x)(x-1)$$

$$\frac{dy}{dx} = 3(1-x)(x-1)$$

$$\frac{dy}{dx} = 3(1-x)(x-1)$$

$$iii) \frac{dy}{dx} = 3(1-x)(x+1)$$

$$0 = 3(1-x)(x+1)$$

$$x = 1 \text{ or } -1$$

When $x = 1, y = 0$ (from i)

When $x = -1, y = -(-1+2)(-1-1)^2$

$$y = -4$$

(1, 0) and (-1, -4)

Question 14

a) Method A: Listing elements of each set

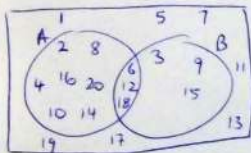
$$A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

$$B = \{3, 6, 9, 12, 15, 18\}$$

$$A \cap B = \{6, 12, 18\}$$

$$A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$$

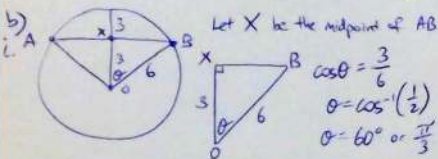
Method B: Venn Diagram



$$i. \frac{10}{20} = \frac{1}{2} \quad ii. \frac{6}{20} = \frac{3}{10}$$

$$iii. \frac{3}{20} \text{ or } P(A) \times P(B) \quad iv. \frac{13}{20} \text{ or } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = \frac{1}{2} + \frac{3}{10} - \frac{3}{20} = \frac{13}{20}$$



$$\therefore \angle AOB = 2 \times 60^\circ \text{ or } 2 \times \frac{\pi}{3}$$

$$= 120^\circ$$

$$= 120 \times \frac{\pi}{180} = \frac{2\pi}{3}$$

ii. Area of minor segment = Area of minor sector - triangle

Area of minor sector

$$A = \frac{1}{2} \times 6^2 \times \frac{2\pi}{3}$$

$$A = 18\pi \text{ cm}^2 \text{ (or } 37.699 \dots)$$

Area of triangle

$$A = \frac{1}{2} \times 6 \times 6 \times \sin\left(\frac{2\pi}{3}\right)$$

$$A = 9\sqrt{3} \text{ cm}^2 \text{ (or } 15.588 \dots)$$

$\therefore$ Area of minor segment

$$18\pi - 9\sqrt{3}$$

$$= 22.11 \text{ cm}^2$$

c) i. Horizontal tangents occur when $\frac{dy}{dx} = 0$

$$\therefore x = 0, 2, 4$$

ii. A function is decreasing when the gradient ($f'(x)$) is negative.

$$\therefore x < 0, 2 < x < 4$$

$$(-\infty, 0) \cup (2, 4)$$

$$d) (\operatorname{cosec} x + \frac{\cos x}{\sin x})(\operatorname{cosec} x - \frac{\cos x}{\sin x})$$

Method A

$$LHS = (\operatorname{cosec} x + \cot x)(\operatorname{cosec} x - \cot x)$$

$$= \operatorname{cosec}^2 x - \cot^2 x$$

$$= 1 = RHS \quad (\text{rearrangement of } 1 + \cot^2 x = \operatorname{cosec}^2 x)$$

Method B

$$LHS = \left(\frac{1}{\sin x} + \frac{\cos x}{\sin x}\right)\left(\frac{1}{\sin x} - \frac{\cos x}{\sin x}\right)$$

$$= \left(\frac{1 + \cos x}{\sin x}\right)\left(\frac{1 - \cos x}{\sin x}\right)$$

$$= \frac{1 - \cos^2 x}{\sin^2 x}$$

$$= \frac{\sin^2 x}{\sin^2 x}$$

$$= 1 = RHS$$

$$e) i. h(x) = f(g(x))$$

$$= \ln(x^4 + 1)$$

ii. Domain

For $\ln A$, $A > 0$.

Since $x^4 + 1 \geq 1$, $\ln(x^4 + 1)$ is valid for all values of x .

$$\therefore x \in \mathbb{R}, (-\infty, \infty)$$

Range

Since $x^4 + 1 \geq 1$, the minimum value of $\ln(x^4 + 1)$ is $\ln 1 = 0$.

$$\therefore h(x) \geq 0 \text{ or } [0, \infty)$$

iii. $f(x)$ is neither $g(x)$ is even $h(x)$ is even

iv. Since $h(x)$ is even, $h(x) = h(-x)$

$$LHS = h(x) + h(-x)$$

$$= \ln(x^4 + 1) + \ln(x^4 + 1)$$

$$= 2 \ln(x^4 + 1)$$

$$= 2 \ln(x^4 + 1)$$

$$= \ln(x^4 + 1)^2 = \ln(g(x)^2)$$

$$= f(g(x)^2) = RHS$$

$$v. i(x) = g(f(x))$$

$$= (\ln x)^4 + 1$$

$$(\ln x)^4 + 1 \neq 4 \ln x + 1$$

Since

$$\ln(x^4) + 1 \leq 4 \ln x + 1$$

$$(\ln x)^4 \geq 0$$

$$\text{so } (\ln x)^4 + 1 \geq 1$$

$$\text{Range: } i(x) \geq 1$$

$$[1, \infty)$$